THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY (GUANGZHOU)

DSAA2011 Machine Learning

L1 Logistics and Introduction

Dr. Zixin Zhong

Data Science and Analytics Thrust
Information Hub
Hong Kong University of Science and Technology (Guangzhou)

January 16, 2026

Outline

1. Logistics
2. Introduction to Machine Learning
   - Definition and taxonomy
3. Mathematical tools
   - Set and function
   - Probability and estimation
   - Systems of linear equations



Course information



Weikai Yang
Assistant Professor

Visual analytics +
Machine learning

W1-316
weikaiyang@hkust-gz.edu.cn



Zixin Zhong
Assistant Professor

Reinforcement learning
Online learning

W1-308
zixinzhong@hkust-gz.edu.cn

- **Lecture time**:
  - [L01] Th 09:00 - 11:50AM Rm 122, E1
  - [L02] Mo 09:00 - 11:50AM Rm 134, E!
- **Lab time**:

  [LA01] Th 07:00 - 07:50PM Rm 122, E1

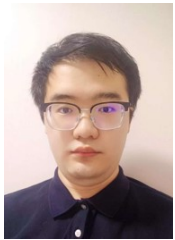
  [LA02] Th 06:00 - 06:50PM Rm 122, E1
- **Course material**:

  Slides, recommended materials
- **Q&A**: Office hour, email, appointment, Teams or Canvas
- **Teaching assistants**:

  Yiyun CHEN, Le CHENG, Na DI,

  Suizi HUANG, Yuhan PU, Jing XU

Meet our TAs



Yiyun CHEN (DSA)



Le CHENG (DSA)



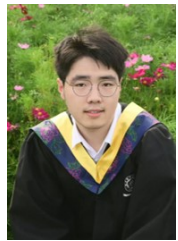
Na DI (DSA)



Suizi HUANG (DSA)



Yuhan PU (DSA)



Jing XU (DSA)



Office hours (weekly, , skipped during PH)

Time	Venue	Instructor/TA	Email
8-9PM Thur	Rm E1-122	Yiyun Chen	ychen780@connect.hkust-gz.edu.cn
		Le Cheng	lcheng916@connect.hkust-gz.edu.cn
		Na Di	ndi595@connect.hkust-gz.edu.cn
		Suizi Huang	shuang310@connect.hkust-gz.edu.cn
		Yuhan Pu	ypu159@connect.hkust-gz.edu.cn
		Jing Xu	jxu289@connect.hkust-gz.edu.cn
7-8PM Tue	Rm W1-316	Weikai Yang	weikaiyang@hkust-gz.edu.cn
	Rm W1-308	Zixin Zhong	zixinzhong@hkust-gz.edu.cn

♡ **Please show respect and appreciation to our TAs!**

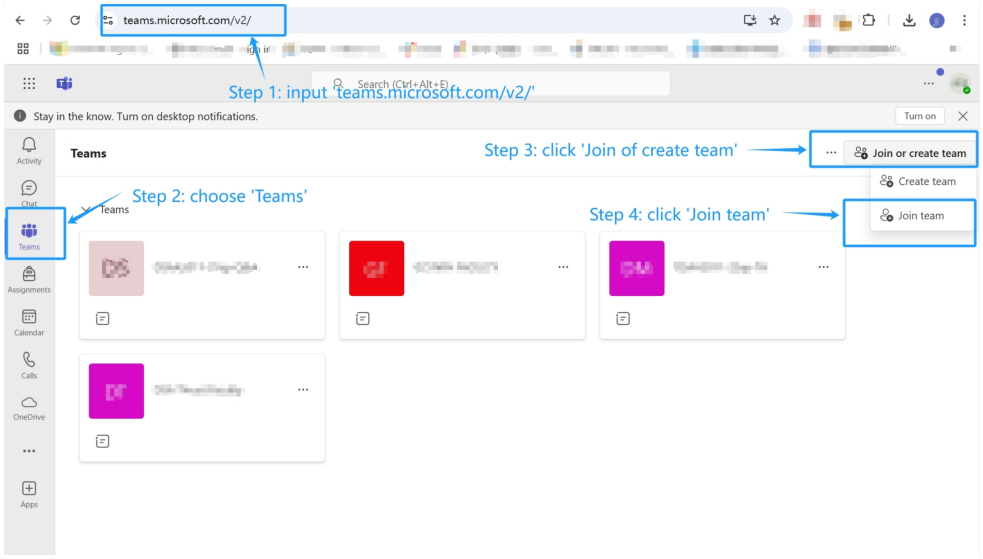


Q&A: Microsoft Teams

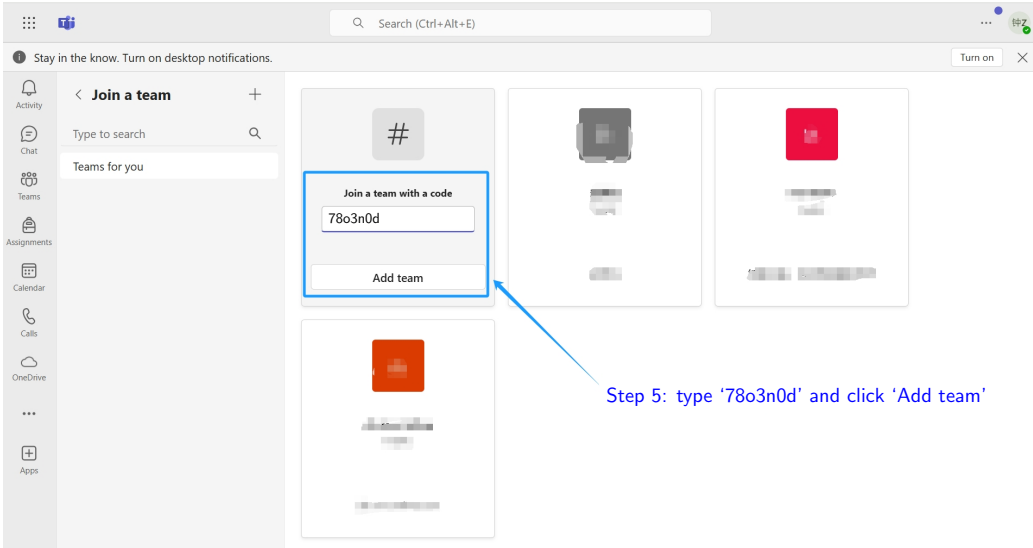
Microsoft Teams (code: 78o3n0d)



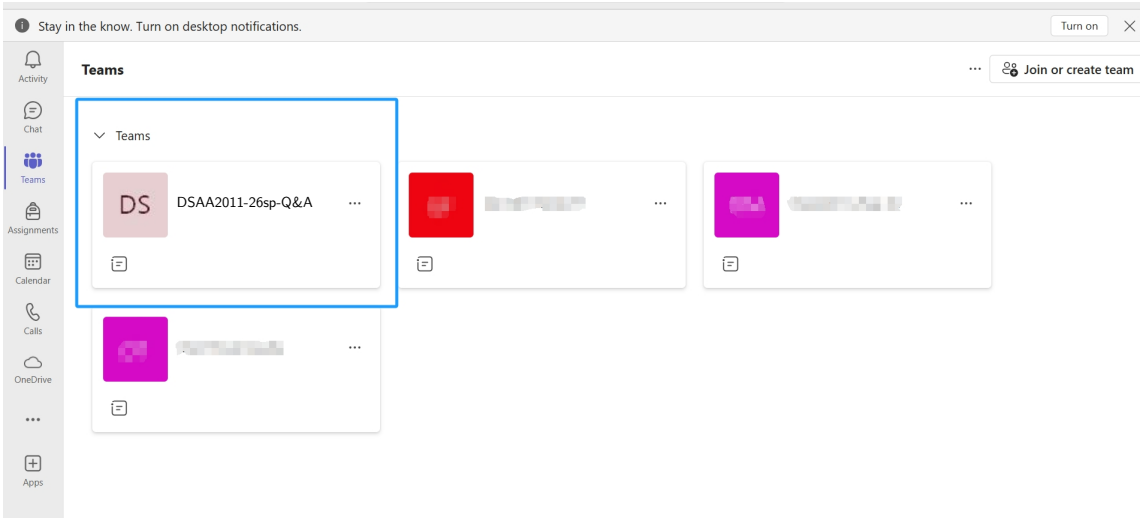
Q&A via Microsoft Teams: https://teams.microsoft.com/v2/



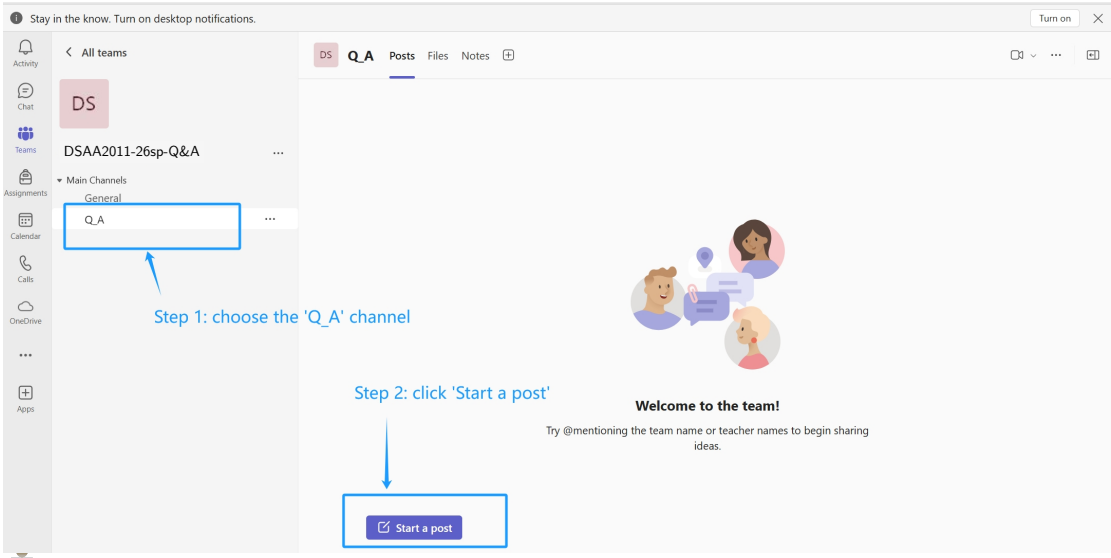
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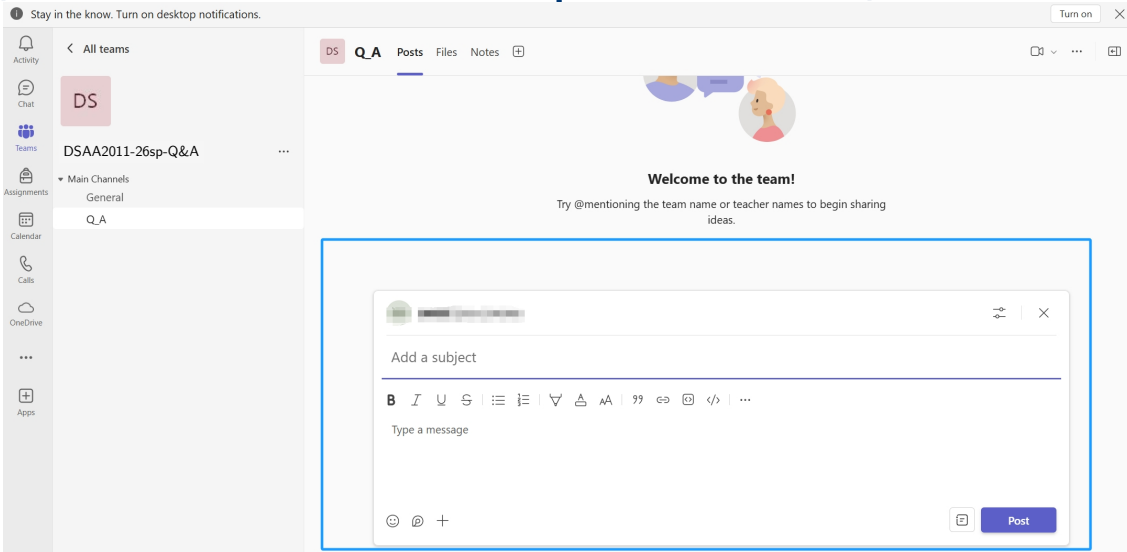
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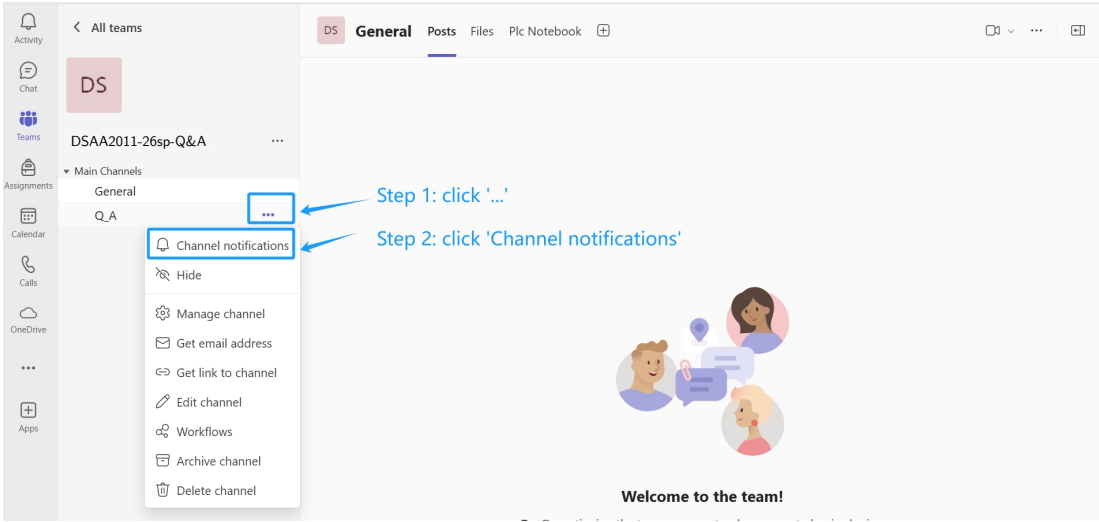
Q&A via Microsoft Teams: ask questions in the 'Q_A' channel



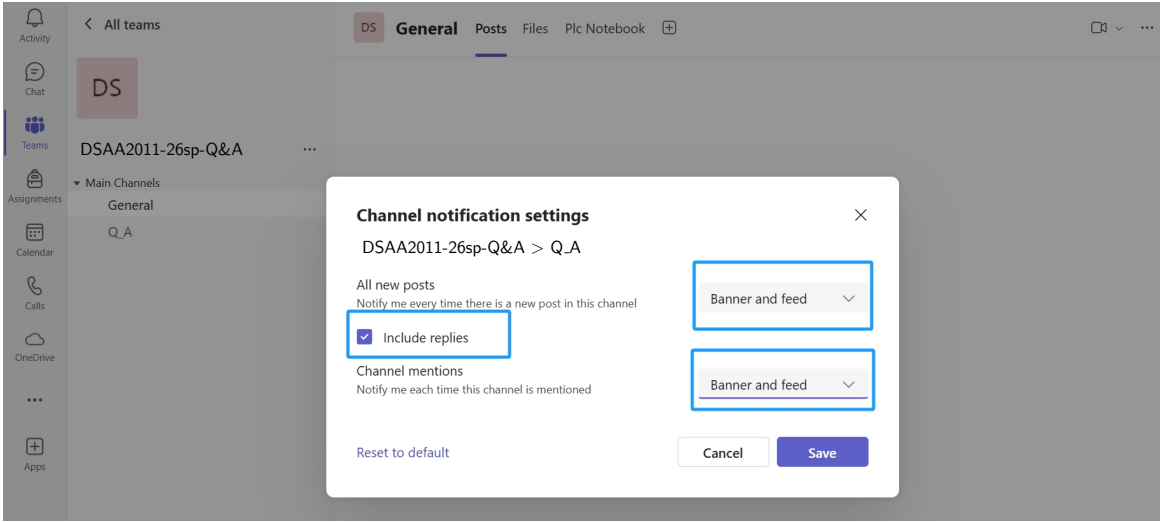
Q&A via Microsoft Teams: ask questions in the ‘Q_A’ channel



Q&A via Microsoft Teams: notification on



Q&A via Microsoft Teams: notification on



Q&A via Microsoft Teams: notification on

Say ‘Hi’ and send blessings to your peers in the ‘Q_A’ channel



Try Canvas: https://hkust-gz.instructure.com/courses/

- Quizzes:
  - Mark midterm time in your calendar!
- Syllabus
- Announcements, Modules
- Discussions, Chat
- Grades





Grading scheme

- **Midterm 20 Points:** 28 March (Sat of Week # 7)
- **Written assignments 40 Points:** group project
- **Final exam 40 Points:** [TBD]
- **Bonus 5 points:**
  - 3 points: proposing good questions/answers, spotting nontrivial mistakes in slides, etc.
  - 2 points: 1 point for 5 lab attendance.



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  - 3 points: proposing good questions/answers, spotting nontrivial mistakes in slides, etc.
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Letter Grade	Raw Mark Range (Reference)
A	$[85, 100]$
B	$[70, 85)$
C	$[55, 70)$
D	$[40, 55)$
F	< 40



Exams and assignment

- Midterm: L01-06
- Final: All materials
- Permitted calculators: Basic non-programmable models only. Devices must not have graphing, CAS, or programmable features.
- No cheet sheet.

- More details will be released later



Exercise in lecture/lab

- Exercise in lab
  - You are highly encouraged to attend lab sessions!
- In-class written exercise
  - Please always **bring a few paper and a pen** for writing exercise during the lecture session.
  - There will be written exercise during lectures.

⊙ Your grade for this course will not be affected.



Syllabus

Lec #	Topic	Lecturer
1	**Introduction + course Info**	Zixin
2-4	Supervised learning: regression and classification	Zixin
5	SVM (supervised learning) + Feature engineering	Zixin
6	Model evaluation and choice	Zixin
7	Ensemble learning (supervised learning) + Feature selection	Weikai
	Midterm (28 March): mark it on calendar!	
8-9	Unsupervised learning: clustering	Weikai
10	Dimension reduction	Weikai
11	Markov and graphical models	Weikai
12	Active learning + Reinforcement learning	Weikai
13	Project presentation	Both
	Project report submission (TBD)	
	Final exam (TBD)	



Academic calendar

Week	S	M	T	W	T	F	S
	January, 2026						
					1	2	3
	4	5	6	7	8	**9**	10
	11	12	13	14	**15**	16	17
1	18	19	20	21	**22**	**23**	**24**
1	25	**26**	**27**	**28**	**29**	**30**	**31**
	February						
2	1	**2**	**3**	**4**	**5**	**6**	**7**
3	8	**9**	**10**	**11**	**12**	**13**	**14**
	15	16	17	18	19	20	21
3	22	23	**24**	**25**	**26**	**27**	**28**
	March						
4	1	**2**	**3**	**4**	5	6	7
5	8	9	10	11	12	13	14
6	15	16	17	18	19	20	21
7	22	23	24	25	26	27	28
8	29	30	31				
	April						
8				1	2	3	4
9	5	6	7	8	9	10	11
10	12	13	14	15	16	17	18
11	19	20	21	22	23	24	25
12	26	27	28	29	30		
	May						
						1	2
13	3	4	5	6	7	8	9
13	10	**11**	**12**	**13**	**14**	**15**	**16**
	17	**18**	**19**	**20**	**21**	**22**	**23**
	24	**25**	**26**	**27**	**28**	**29**	30
	31						

Events	
9	Validation Period Starts - All UG/PG Students
15	Class Enrollment Starts - All UG/PG Students
22	Last day of Winter Term
23	Spring Term commences
23/1-5/2	Add/Drop Period
11-23	Winter Break for UG students (no centrally scheduled PG classes)
28	Make-up day for Monday classes
6/2-4/3	Extended Drop Period - For PG Courses Only
11	Last day of Spring Term classes
12-16	Study Break
18-29	Spring Term Examinations
29	Last day of Spring Term

General Holidays	
1-3	*New Year's Day*
15-23	*Spring Festival*
4-6	*Qingming Festival*
1-5	*Labor Day*



Programming language policy

- We will do demos in **Python (Jupyter Lab)**.
- You're welcome to use any language you like (that your TAs can read) for homework or project.
- TAs will only support **Python**.



Collaboration policy

Lecture note/Project: yes of course! Clarify the contributions of each member.

- Give credit to the people who have helped you: write on your report the names of the people you worked with.
- Give credit to the other resources that have helped you: please write on your report the GenAI tools, textbooks, notes, or web pages you found useful.

Midterm/Final: Figure out your solutions by yourself.

- All of the text that you submit should be done by you.
- No collaboration/discussion here!



IP policy

Other course note websites:

- do not post any course materials in other websites. this makes the next rendition of the course worse for everyone.
- please report to us any course materials you find online (not on our websites).



How to do well in this course?

- Quick look at lecture slides and lab notes in advance
- Pay attention to lecture and lab
- Ask **whenever**
- If you fail to follow, ask for a pause or come back later
- **No/minimum use of electronic devices** during lecture (use them only for participating in online quizzes)



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  - Treat each lab note as a quiz (good practise)
- Practice all you learn in the project



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- Do exercise in lecture notes and lab notes
  - Treat each lab note as a quiz (good practise)
- Practice all you learn in the project
- Find a peer, describe a concept or teach each other
- Tell us how to improve this course



How to do well in this course?

- Be perserverant:

  "The brick walls are there for a reason. The brick walls are not there to keep us out. The brick walls are there to give us a chance to show how badly we want something. Because the brick walls are there to stop the people who don't want it badly enough. They're there to stop the other people."

  from The Last Lecture by Randy Pausch



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- Brain power is like muscle power. It can only get better through heavy lifting. Everyone can become smarter by training (practicing, analyzing, coding, ...)



Memorize everything: formula, equation, proof ...?

Understand the idea of algorithm design
See the beauty and importance of math in ML



Outline

1. Logistics

2. Introduction to Machine Learning
   - Definition and taxonomy

3. Mathematical tools
   - Set and function
   - Probability and estimation
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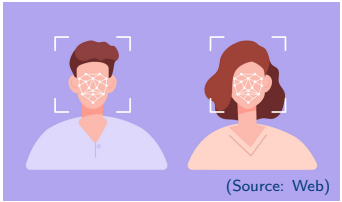


Machine Learning is everywhere



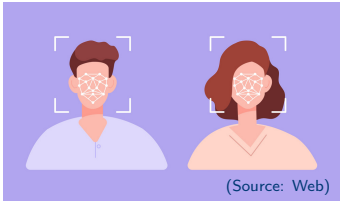


Machine Learning is everywhere



Identity verification

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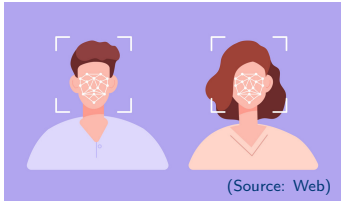


Identity verification



Large Language Model

Machine Learning is everywhere



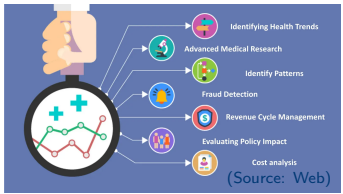
Identity verification

香港科技大学(广州)
THE HONG KONG
UNIVERSITY OF SCIENCE AND
TECHNOLOGY (GUANGZHOU)

HKUST(GZ) GPT Service System

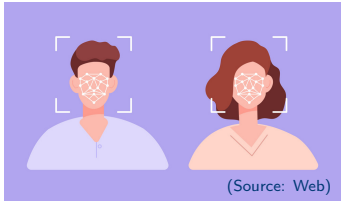
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Large Language Model



Healthcare

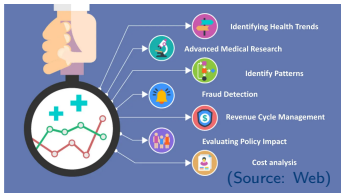
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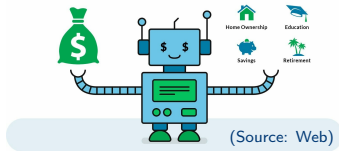
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Large Language Model



Healthcare



Wealth management

What is Machine Learning (ML)?



Definition of Machine Learning

Prof. Tom M. Mitchell from CMU (Samuel, 1959):

- A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P, if its performance at tasks in T, as measured by P, improves with experience E.



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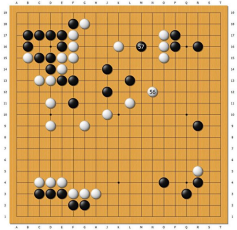
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(Source: Web)

- Experience E (data): games played by the program or human



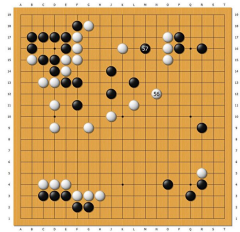
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- Experience E (data): games played by the program or human
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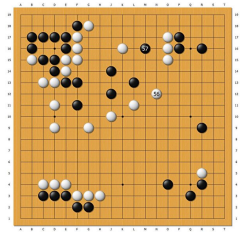
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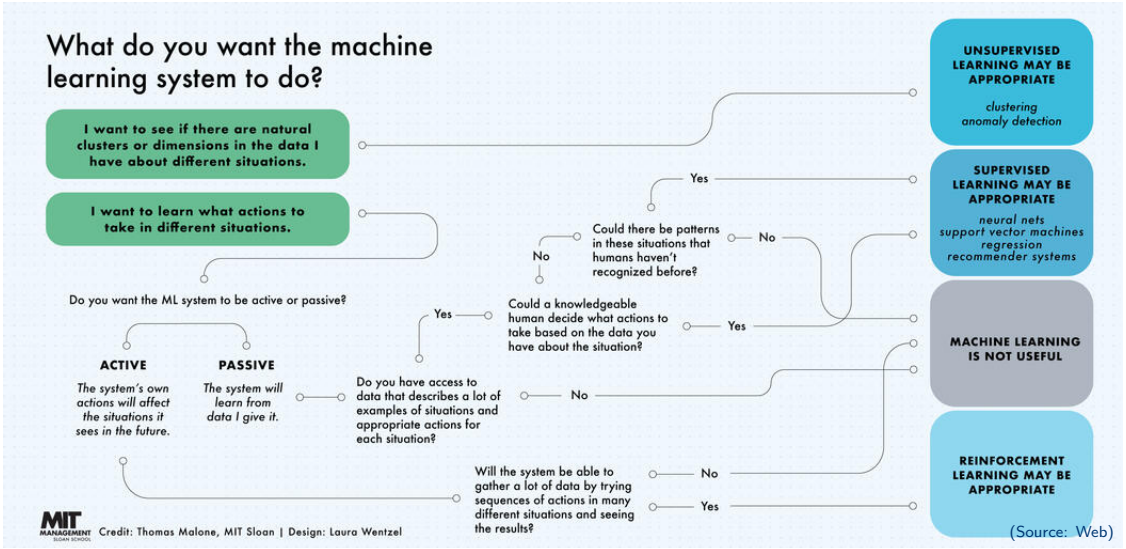
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- Experience E (data): games played by the program or human
- Performance measure P: winning rate
- Task T: to win



Taxonomy of Machine Learning



(Source: Web)

Taxonomy of Machine Learning

Supervised Learning	Unsupervised Learning	Reinforcement Learning
Has outcome information (“labels”)	No outcome information available	Makes decisions based on trial and error
Finds patterns that relate to those outcomes	Analyzes or identifies groups without labels or human instruction	Decision-making algorithm is constantly refined based on “rewards”
Uses patterns to predict outcomes not yet known	Offers insights into characteristics that define groups	Excels in complex situations

Pecan

(Source: Web)



Taxonomy of Machine Learning (A Simplistic View)

- What type of data?
  - **Supervised learning** - labeled data: e.g., prediction
  - Semi-supervised learning
  - **Unsupervised learning** - unlabeled data: e.g., clustering



Taxonomy of Machine Learning (A Simplistic View)

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Taxonomy of Machine Learning (A Simplistic View)

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  - Semi-supervised learning
  - **Unsupervised learning** - unlabeled data: e.g., clustering
  - **Reinforcement learning** - environment feedback: e.g., multi-armed bandit

- When do we collect data?
  - **Offline learning**
  - **Online learning**



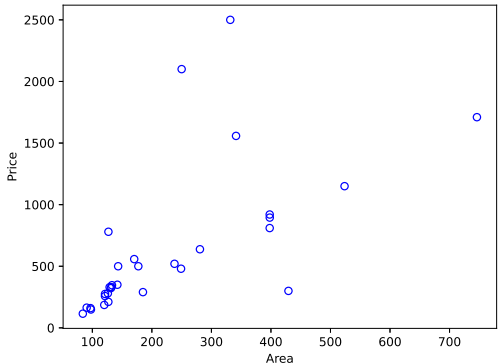
Supervised Learning



Task: housing price prediction

- Given a dataset containing n samples

$$\left(x^{(1)}, y^{(1)}\right), \left(x^{(2)}, y^{(2)}\right), \left(x^{(3)}, y^{(3)}\right), \ldots, \left(x^{(n)}, y^{(n)}\right)$$



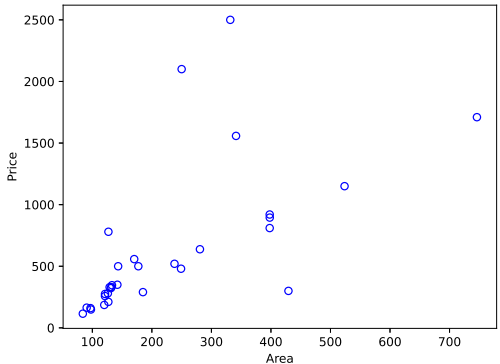


Task: housing price prediction

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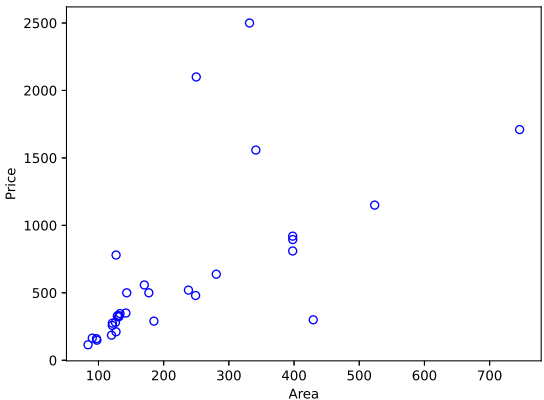
- Task: if a residence has $x = 320$ square kilometers, can we predict its price?





Task: housing price prediction

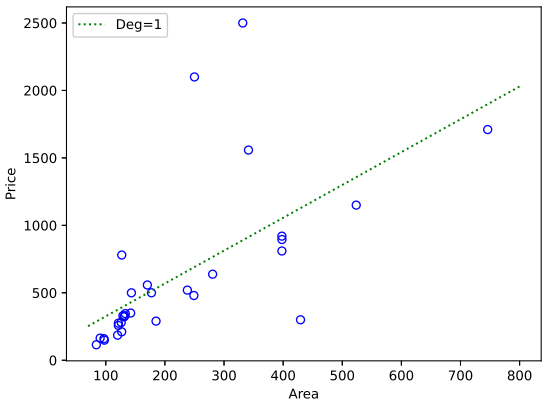
- Linear regression (polynomial regression with degree 1):
  Assume $y = p_1(x) = ax + b$ and guess the values of a and b.





Task: housing price prediction

- Linear regression (polynomial regression with degree 1):
  Assume $y = p_1(x) = ax + b$ and guess the values of a and b.



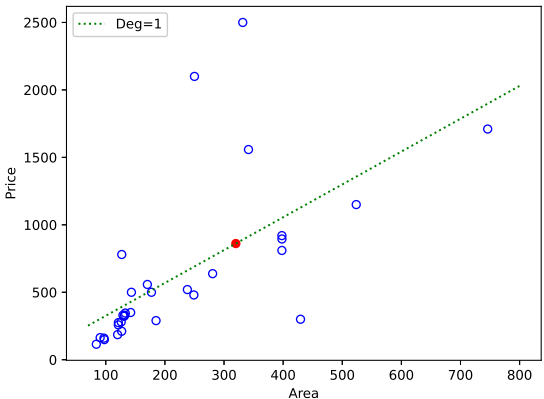
$a = 2.43, b = 82.36$

$p_1(x) = 2.43x + 82.36$



Task: housing price prediction

- Linear regression (polynomial regression with degree 1):
  Assume $y = p_1(x) = ax + b$ and guess the values of a and b.



$a = 2.43, b = 82.36$

$p_1(x) = 2.43x + 82.36$

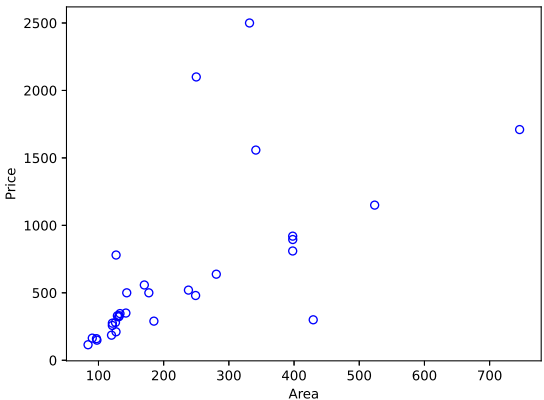
$p_1(320) = 861.29$

- If a residence has $x = 320$ square kilometers, we predict its price as $y = 861.29$ CNY.



Task: housing price prediction

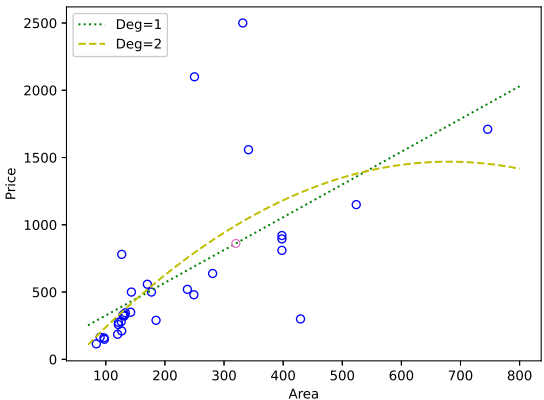
- Polynomial regression with degree 2:
  Assume $y = p_2(x) = ax^2 + bx + c$ and guess the values of a, b and c.





Task: housing price prediction

- Polynomial regression with degree 2:
  Assume $y = p_2(x) = ax^2 + bx + c$ and guess the values of a, b and c.



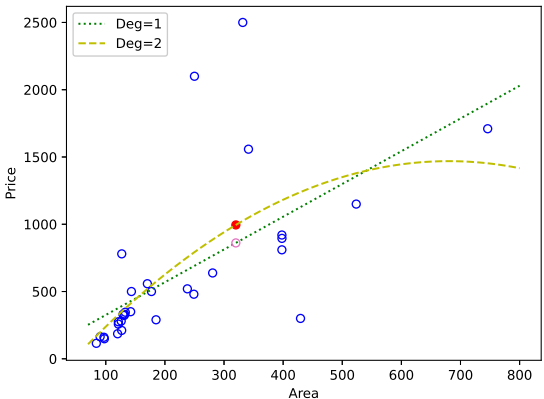
$a = -0.0037, b = 4.97, c = -221.8$

$p_2(x) = -0.0037x^2 + 4.97x - 221.8$



Task: housing price prediction

- Polynomial regression with degree 2:
  Assume $y = p_2(x) = ax^2 + bx + c$ and guess the values of a, b and c.



$a = -0.0037, b = 4.97, c = -221.8$

$p_2(x) = -0.0037x^2 + 4.97x - 221.8$

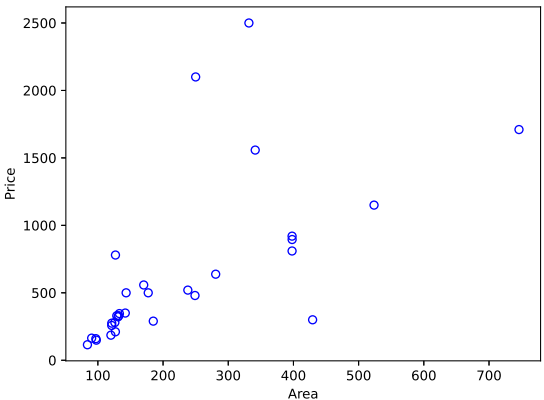
$p_2(320) = 994.94$

- If a residence has $x = 320$ square kilometers, we predict its price as $y = 994.94$ CNY.



Task: housing price prediction

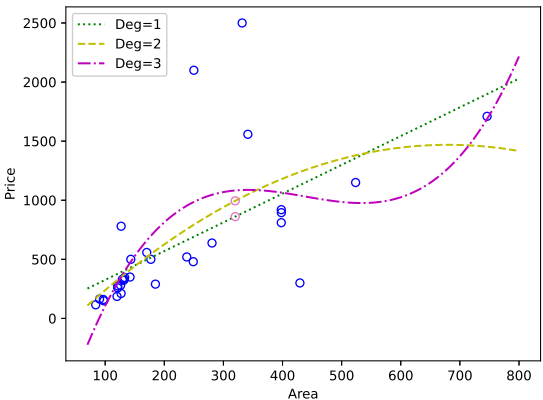
- Polynomial regression with degree 3:
  Assume $y = p_3(x) = ax^3 + bx^2 + cx + d$ and guess the values of a, b, c and d.





Task: housing price prediction

- Polynomial regression with degree 3:
  Assume $y = p_3(x) = ax^3 + bx^2 + cx + d$ and guess the values of a, b, c and d.



$a = 3.16 * 10^{-5}, b = -0.042,$

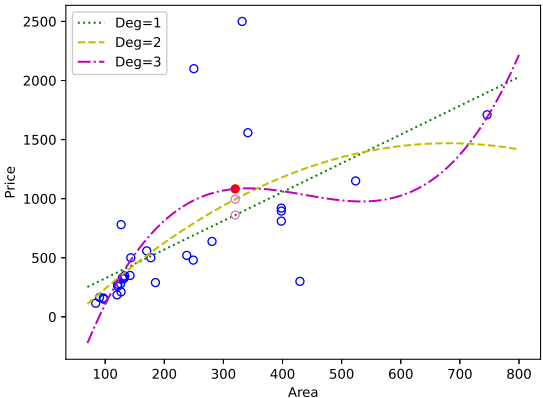
$c = 17.33, d = -1243$

$p_3(x) = 3.16 * 10^{-5}x - 0.042x + 17.33x - 1243$



Task: housing price prediction

- Polynomial regression with degree 3:
  Assume $y = p_3(x) = ax^3 + bx^2 + cx + d$ and guess the values of a, b, c and d.



$a = 3.16 * 10^{-5}, b = -0.042,$

$c = 17.33, d = -1243$

$p_3(x) = 3.16 * 10^{-5}x - 0.042x + 17.33x - 1243$

$p_3(320) = 1082.80$

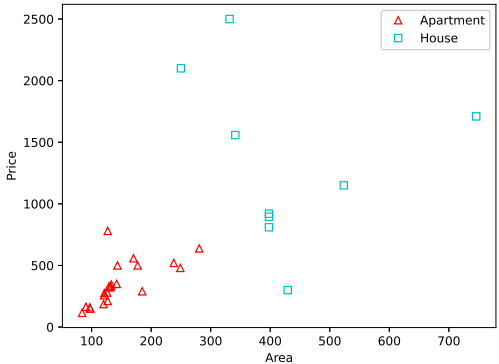
- If a residence has $x = 320$ square kilometers, we predict its price as $y = 1082.80$ CNY.



Task: housing type prediction

- Given a dataset containing n samples ($z \in \{0, 1\}$)

$$\left(x^{(1)}, y^{(1)}, z^{(1)}\right), \left(x^{(2)}, y^{(2)}, z^{(2)}\right), \left(x^{(3)}, y^{(3)}, z^{(3)}\right), \ldots, \left(x^{(n)}, y^{(n)}, z^{(n)}\right)$$

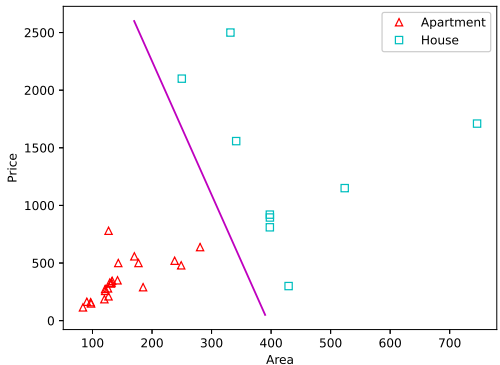


Task: housing type prediction

- Given a dataset containing n samples ($z \in \{0, 1\}$)

$$\left(x^{(1)}, y^{(1)}, z^{(1)}\right), \left(x^{(2)}, y^{(2)}, z^{(2)}\right), \left(x^{(3)}, y^{(3)}, z^{(3)}\right), \ldots, \left(x^{(n)}, y^{(n)}, z^{(n)}\right)$$

- Task: if a residence has $x = 320$ square kilometers and worth $y = 1080.20$ CNY, can we predict its type z?



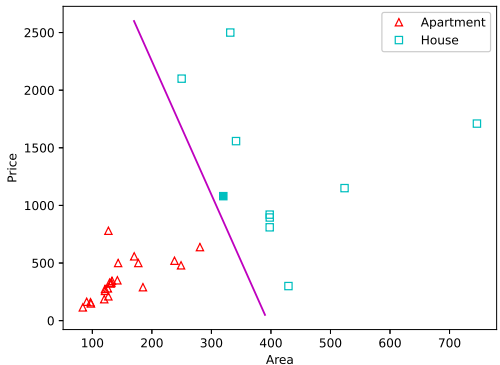


Task: housing type prediction

- Given a dataset containing n samples ($z \in \{0, 1\}$)

$$\left(x^{(1)}, y^{(1)}, z^{(1)}\right), \left(x^{(2)}, y^{(2)}, z^{(2)}\right), \left(x^{(3)}, y^{(3)}, z^{(3)}\right), \ldots, \left(x^{(n)}, y^{(n)}, z^{(n)}\right)$$

- Task: if a residence has $x = 320$ square kilometers and worth $y = 1080.20$ CNY, can we predict its type z?





Task comparison: regression vs classification

- **Regression**: the prediction result is a continuous variable
  - e.g., price prediction
    $(x = \text{Area}) \rightarrow (y = \text{Price})$?
- **Classification**: the prediction result is a discrete variable
  - e.g., type prediction
    $(x = \text{Area}, y = \text{Price}) \rightarrow (z = \text{Type})$?



Unsupervised Learning

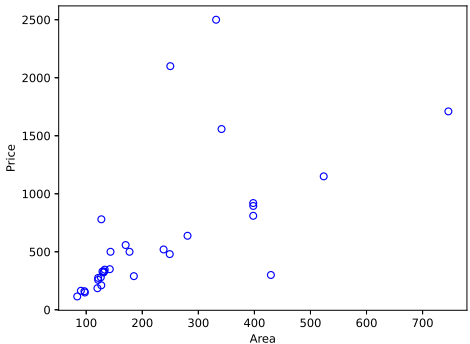


Task: housing type prediction

- Given a dataset containing n samples

$$\left(x^{(1)}, y^{(1)}\right), \left(x^{(2)}, y^{(2)}\right), \left(x^{(3)}, y^{(3)}\right), \ldots, \left(x^{(n)}, y^{(n)}\right)$$

- Task: can we learn the type of each residence?



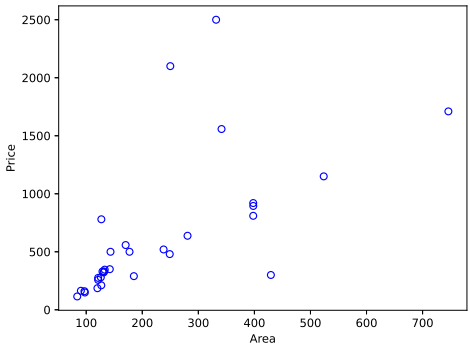


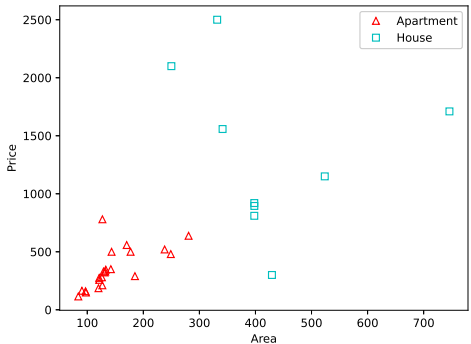
Task: housing type prediction

- Given a dataset containing n samples

$$\left(x^{(1)}, y^{(1)}\right), \left(x^{(2)}, y^{(2)}\right), \left(x^{(3)}, y^{(3)}\right), \ldots, \left(x^{(n)}, y^{(n)}\right)$$

- Task: can we learn the type of each residence?





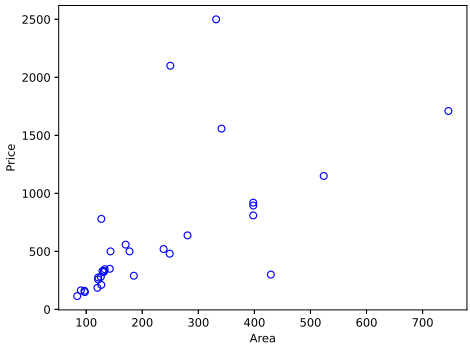


Clustering

- Given a dataset containing n samples

$$\left(x^{(1)}, y^{(1)}\right), \left(x^{(2)}, y^{(2)}\right), \left(x^{(3)}, y^{(3)}\right), \ldots, \left(x^{(n)}, y^{(n)}\right)$$

- Task (vague): find interesting structures in the data



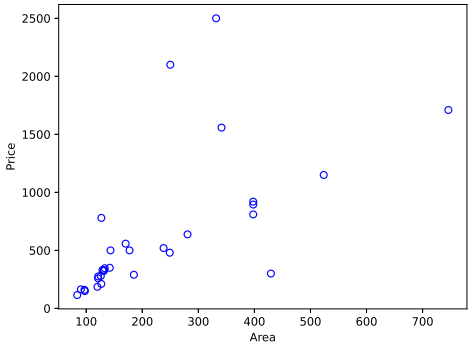


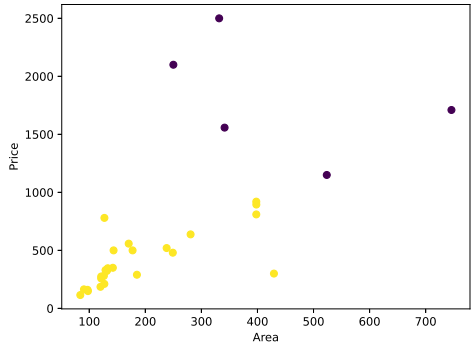
Clustering

- Given a dataset containing n samples

$$\left(x^{(1)}, y^{(1)}\right), \left(x^{(2)}, y^{(2)}\right), \left(x^{(3)}, y^{(3)}\right), \ldots, \left(x^{(n)}, y^{(n)}\right)$$

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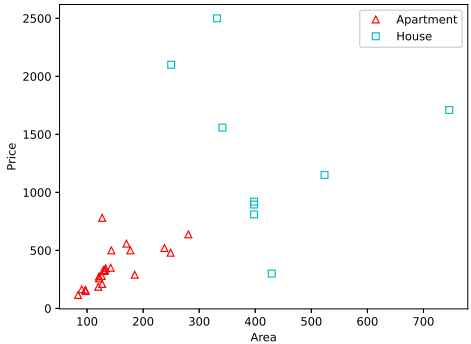


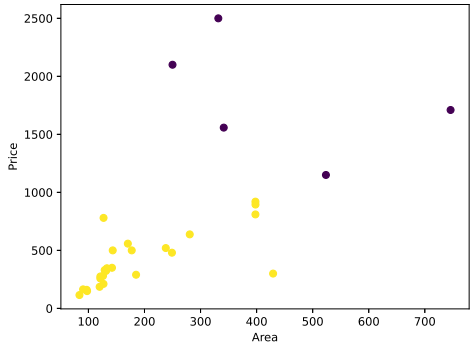
Clustering

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$$\left(x^{(1)}, y^{(1)}\right), \left(x^{(2)}, y^{(2)}\right), \left(x^{(3)}, y^{(3)}\right), \ldots, \left(x^{(n)}, y^{(n)}\right)$$

- Task (vague): find interesting structures in the data





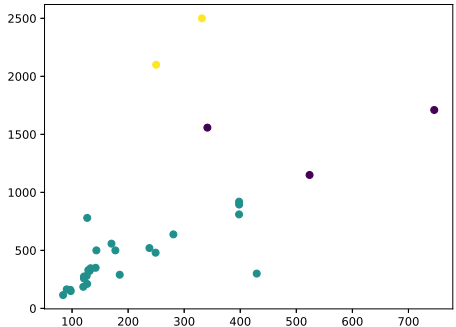


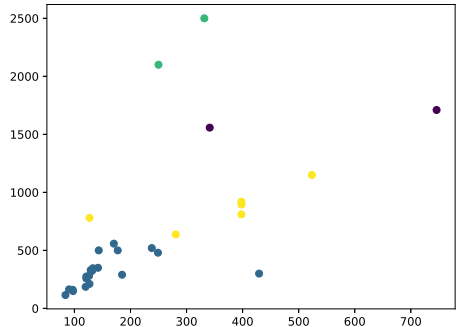
Clustering

- Given a dataset containing n samples

$$\left(x^{(1)}, y^{(1)}\right), \left(x^{(2)}, y^{(2)}\right), \left(x^{(3)}, y^{(3)}\right), \ldots, \left(x^{(n)}, y^{(n)}\right)$$

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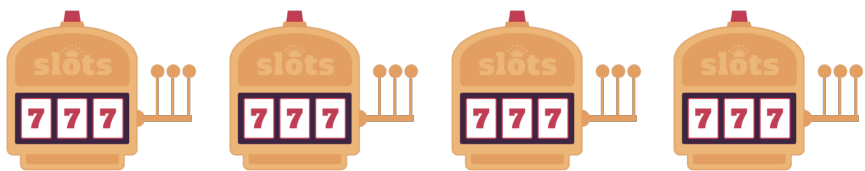
Reinforcement Learning



Task: win more

- We pull one of the 4 arms of a slot machine at each round t

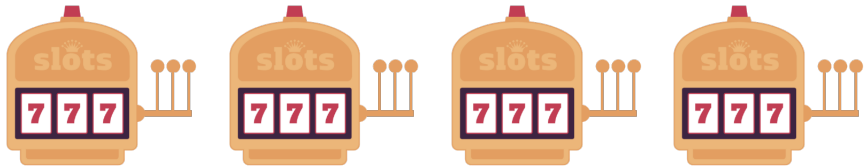
$L = 4$





Task: win more

- We pull one of the 4 arms of a slot machine at each round t
- Each arm i yields a random reward $W_t(i)$ at round t



$L = 4$

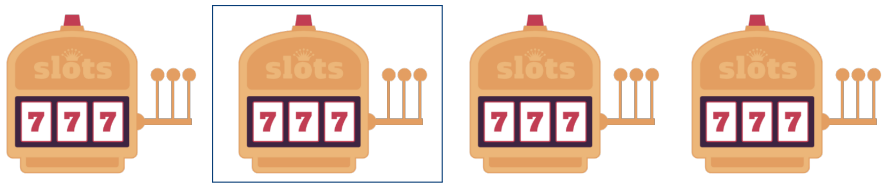
Reward $W_t(1)$ $W_t(2)$ $W_t(3)$ $W_t(4)$



Task: win more

- We pull one of the 4 arms of a slot machine at each round t
- Each arm i yields a random reward $W_t(i)$ at round t
- At time t, if we pull arm 2, we earn $W_t(2)$

$L = 4$



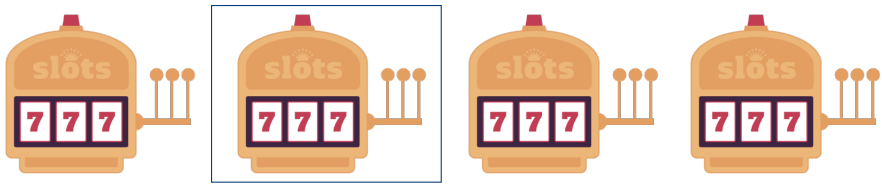
Reward	$W_t(1)$	$\boxed{W_t(2)}$	$W_t(3)$	$W_t(4)$



Task: win more

- We pull one of the 4 arms of a slot machine at each round t
- Each arm i yields a random reward $W_t(i)$ at round t
- At time t, if we pull arm 2, we earn $W_t(2)$
- Task: how can we earn as much as possible after 100 rounds?

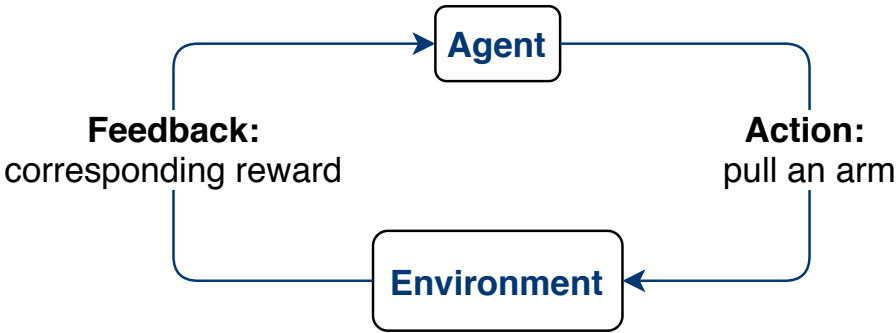
$L = 4$



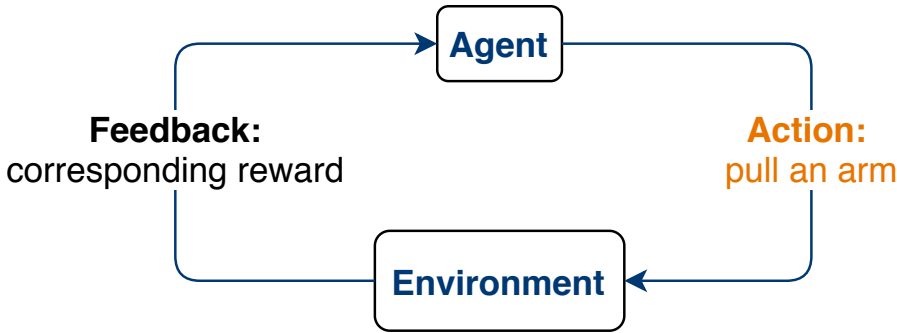
Reward	$W_t(1)$	$W_t(2)$	$W_t(3)$	$W_t(4)$



Multi-armed bandit problem (MAB)



Multi-armed bandit problem (MAB)



- Instead of labeled/unlabeled data, a reinforcement learning algorithm learns from environment feedback.



Taxonomy of Machine Learning (A Simplistic View)

- What type of data?
  - **Supervised learning** - labeled data: e.g., prediction
  - Semi-supervised learning
  - **Unsupervised learning** - unlabeled data: e.g., clustering
  - **Reinforcement learning** - environment feedback: e.g., multi-armed bandit

- When do we collect data?
  - **Offline learning**
  - **Online learning**



Outline

1. Logistics
2. Introduction to Machine Learning
   - Definition and taxonomy
3. Mathematical tools
   - Set and function
   - Probability and estimation
   - Systems of linear equations



Sets

- A set S is an unordered collection of objects.
- $S = \{1, 2, 3, 4, 5, 6\}$ is the possible outcomes of the toss of a die.
- $S = [a, b] = \{x \in \mathbb{R} : a \leq x \leq b\}$ is the set of all numbers from a to b inclusive.
- $S = (a, b] = \{x \in \mathbb{R} : a < x \leq b\}$ is the set of all numbers from a to b, excluding a including b.
- $\mathbb{R}$ is the set of all real numbers.
- $\mathbb{R}^d$ is the set of all real vectors of length d.

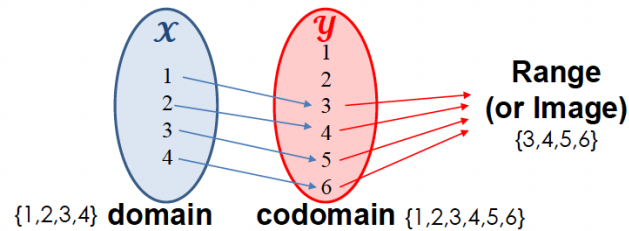


Functions

- A function f is a map from a set X to another set Y. We write this as

$$f : X \to Y.$$

- For example, the function $f : \mathbb{R} \to [0, \infty)$ could be given by the recipe $f(x) = x^2$.
- The set of inputs is called the domain; the set of possible outputs is called the codomain; the set $\{f(x) : x \in X\}$ is called the range (or image).
- For example $f : \{1, 2, 3, 4\} \to \{1, 2, 3, 4, 5, 6\}$ given by the recipe $f(x) = x + 2$ has codomain $\{1, 2, 3, 4, 5, 6\}$ and range $\{3, 4, 5, 6\}$.



Functions

Definition 3.1

Let $f(x)$ be a continuous function defined on domain $\mathbb{R}$.

- $f(x)$ is **differentiable** if $f'(x) = \lim_{x \to c} \frac{f(x)-f(c)}{x-c}$ exists for all $x \in \mathbb{R}$.
- $f(x)$ is **convex** if $f(\lambda a + (1-\lambda)b) \leq \lambda f(a) + (1-\lambda)f(b)$ for all $a, b \in \mathbb{R}$;
- $f(x)$ is **strictly convex** if $f(\lambda a + (1-\lambda)b) < \lambda f(a) + (1-\lambda)f(b)$ for all $a, b \in \mathbb{R}$;
- $f(x)$ is **concave** if $f(\lambda a + (1-\lambda)b) \geq \lambda f(a) + (1-\lambda)f(b)$ for all $a, b \in \mathbb{R}$;
- $f(x)$ is **strictly concave** if $f(\lambda a + (1-\lambda)b) > \lambda f(a) + (1-\lambda)f(b)$ for all $a, b \in \mathbb{R}$;



Functions

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- $f(x)$ is **strictly concave** if $f(\lambda a + (1-\lambda)b) > \lambda f(a) + (1-\lambda)f(b)$ for all $a, b \in \mathbb{R}$;

- $f(x)$ is differentiable and convex: x_0 such that $f'(x_0) = 0$ minimize function f.
- $f(x)$ is differentiable and concave: x_0 such that $f'(x_0) = 0$ maximize function f.



Linear functions

A function $f : \mathbb{R}^d \to \mathbb{R}$ is linear if it satisfies

- **(Homogeneity)** For any vector $\mathbf{x} \in \mathbb{R}^d$ and scalar $a \in \mathbb{R}$,

$$f(a\,\mathbf{x}) = a\,f(\mathbf{x})$$

- **(Additivity)** For any two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$,

$$f(\mathbf{x} + \mathbf{y}) = f(\mathbf{x}) + f(\mathbf{y})$$

Note that a linear function $f : \mathbb{R}^d \to \mathbb{R}$ must pass through the origin, i.e., $f(\mathbf{0}) = 0$ where $\mathbf{0} \in \mathbb{R}^d$ is the zero vector in d dimensions. Why?



Affine functions

- An affine function f is a linear function plus possibly a constant.
- More precisely, a function $f : \mathbb{R}^d \to \mathbb{R}$ is affine if it can be expressed as

$$f(\mathbf{x}) = \mathbf{a}^\top \mathbf{x} + b$$

  for some vector $\mathbf{a} \in \mathbb{R}^d$ and some scalar $b \in \mathbb{R}$.
- The scalar b is called the bias or offset.

Example: The following function $f : \mathbb{R}^2 \to \mathbb{R}$ is affine. Why?

$$f(\mathbf{x}) = f(x_1, x_2) = -x_1 + 3x_2 + 7.$$

Exercise: Is a linear function affine? Is an affine function linear?



Linear and Affine Functions

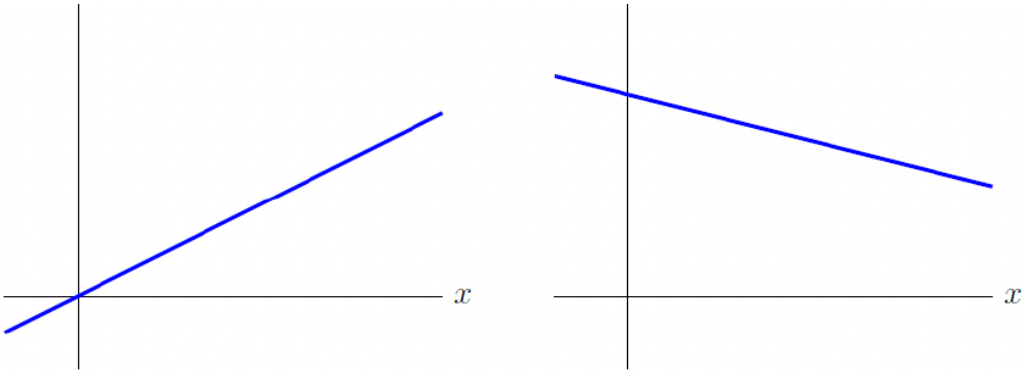


Figure: Left: Linear; Right: Affline but not linear.



Local and global extrema

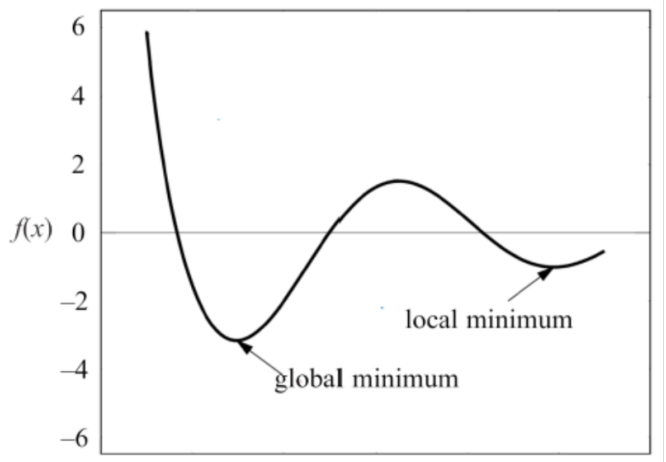
- Consider a function $f : [a, b] \to \mathbb{R}$.
- The function f has a local minimum at $c \in \mathbb{R}$ if $f(x) \geq f(c)$ for all x in an open neighborhood of c.
- The function f has a global minimum at $c \in \mathbb{R}$ if $f(x) \geq f(c)$ for all $x \in [a, b]$.

Exercise: If c is a local minimum of f, is it a global minimum? If c is a global minimum of f, is it a local minimum?

Exercise: How would you define local maximum and global maximum?



Local and global extrema

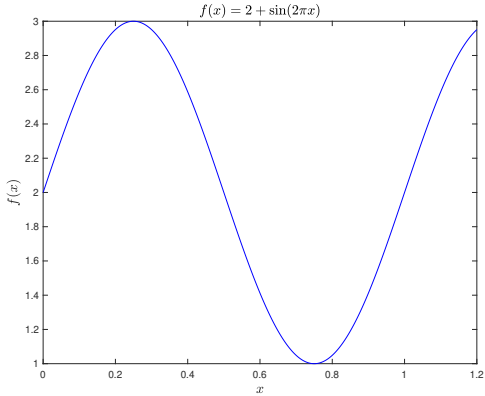


min **and** arg min

- For a function $f : X \to Y$, the minimum $\min_{x \in X} f(x)$ returns the smallest value among all elements in the set $\{f(x) : x \in X\}$.
- For a function $f : X \to Y$, the argmin $x^* = \arg\min_{x \in X} f(x)$ returns the value of $x \in X$ that minimizes $f(x)$, i.e.,
$$f(x^*) = \min_{x \in X} f(x)$$
- arg min returns a value from the domain of the function X and min returns from the range (codomain) Y of the function.
- Let $X = \{0, 1\}$ and $f(0) = \pi$ and $f(1) = e$. Then
$$\arg\min_{x \in X} f(x) = 1 \qquad \min_{x \in X} f(x) = e,$$
and
$$\arg\max_{x \in X} f(x) = 0 \qquad \max_{x \in X} f(x) = \pi.$$



min **and** arg min



- Let $f : X = [0, 1.2] \to \mathbb{R}$ be defined as $f(x) = 2 + \sin(2\pi x)$ (see plot above). Then

$$\arg\min_{x \in X} f(x) = 3/4 \qquad \min_{x \in X} f(x) = +1$$



Derivatives

- For a multivariable function $f : \mathbb{R}^d \to \mathbb{R}$, its gradient vector or derivative at $\mathbf{x} \in \mathbb{R}^d$ is the column vector

$$\nabla_{\mathbf{x}} f(\mathbf{x}) = \frac{\mathrm{d}f}{\mathrm{d}\mathbf{x}} = \begin{bmatrix} \frac{\partial f}{\partial x_1} & \frac{\partial f}{\partial x_2} & \cdots & \frac{\partial f}{\partial x_d} \end{bmatrix}^\top$$

- Recall that $\frac{\partial f}{\partial x_i}$ is the partial derivative of f with respect to the scalar variable x_i.
- For example, if $f(x_1, x_2) = 2x_1^2 + 5x_1x_2 + 3x_2^3$, then

$$\frac{\partial f}{\partial x_1} = 4x_1 + 5x_2 \quad \text{and} \quad \frac{\partial f}{\partial x_2} = 5x_1 + 9x_2^2.$$



Important derivatives

- There are only two derivatives for functions $f : \mathbb{R}^d \to \mathbb{R}$ that take vectors to scalars you need to know for now.
- For a fixed vector $\mathbf{a} \in \mathbb{R}^d$, consider $f(\mathbf{x}) = \mathbf{a}^\top \mathbf{x}$ (known as the inner or dot product). Then
$$\nabla_{\mathbf{x}} f(\mathbf{x}) = \mathbf{a}.$$
- For a fixed matrix $\mathbf{A} \in \mathbb{R}^{d \times d}$, consider $f(\mathbf{x}) = \mathbf{x}^\top \mathbf{A}\mathbf{x}$ (known as the quadratic form). Then
$$\nabla_{\mathbf{x}} f(\mathbf{x}) = (\mathbf{A} + \mathbf{A}^\top)\mathbf{x}.$$
- In most applications, $\mathbf{A}$ is a symmetric matrix (i.e., $\mathbf{A} = \mathbf{A}^\top$) so
$$\nabla_{\mathbf{x}} f(\mathbf{x}) = 2\mathbf{A}\mathbf{x}.$$
- This generalizes the basic fact that if $f(x) = ax^2$, then $\frac{\mathrm{d}f}{\mathrm{d}x} = 2ax$.



Important derivatives

Exercise: Show from the definition of $f(\mathbf{x}) = \mathbf{a}^\top \mathbf{x}$ that

$$\nabla_{\mathbf{x}} f(\mathbf{x}) = \mathbf{a}.$$

Exercise: Show from the definition of $f(\mathbf{x}) = \mathbf{x}^\top \mathbf{A}\mathbf{x}$ that

$$\nabla_{\mathbf{x}} f(\mathbf{x}) = (\mathbf{A} + \mathbf{A}^\top)\mathbf{x}.$$

You'll be forgiven for having to exhibit a substantial amount of meticulous bookkeeping here.

Advice: It is difficult to remember a lot of derivative formulas of complicated multivariate functions. Usually, one consults the Matrix Cookbook
`https://www2.imm.dtu.dk/pubdb/edoc/imm3274.pdf`



Outline

1. Logistics
2. Introduction to Machine Learning
   - Definition and taxonomy
3. Mathematical tools
   - Set and function
   - Probability and estimation
   - Systems of linear equations



Statistical machine learning deals with uncertainty

$\Downarrow$

Definitions of events, probabilities, joint probabilities and conditional probabilities



Notations

- Random variables (r.v.s): upper (e.g., X)
- Values r.v.s take on: lower case (e.g., x, x_i)
- Sets (in which the r.v.s assume their values): calligraphic font (e.g., $\mathcal{X}$)
- Probability, expectation and variance operators: $\Pr(\cdot)$, $\mathbb{E}[\cdot]$, $\mathrm{Var}(\cdot)$



Discrete random variables

Random variables X and Y:

- X can only take values in the finite set $\mathcal{X} = \{x_1, x_2 \ldots, x_M\}$.
- Y can only take values in the finite set $\mathcal{Y} = \{y_1, y_2 \ldots, y_L\}$.

A random variable is said to be **discrete** if the set of values it can take has a finite number of values.

$$\sum_{i=1}^{M} \Pr(X = x_i) = 1, \sum_{i=1}^{L} \Pr(Y = y_i) = 1.$$



Event

Consider m trials (of sampling X and Y) and let the number of trials for which $X = x_i$ and $Y = y_j$ be m_{ij}. Then, if m is large, we can assume that

$$\Pr(X = x_i,\ Y = y_j) = \frac{m_{ij}}{m}.$$

- The argument of $\Pr(\cdot)$ is known as an **event**.
- **Event**: A set in which a r.v. (or multiples r.v.s) assume some value(s) in some set. e.g., $\{X = x_i,\ Y = y_j\}$.



Sum rule and product rule

- **Sum rule**:

$$\Pr(X = x_i) = \sum_{j=1}^{L} \Pr(X = x_i, Y = y_j), \quad \Pr(Y = y_j) = \sum_{i=1}^{M} \Pr(X = x_i, Y = y_j).$$

- **Product rule**:

$$\Pr(X = x_i, Y = y_j) = \Pr(X = x_i | Y = y_j) \cdot \Pr(Y = y_j).$$



Sum rule and product rule

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$$\Pr(X = x_i) = \sum_{j=1}^{L} \Pr(X = x_i, Y = y_j), \quad \Pr(Y = y_j) = \sum_{i=1}^{M} \Pr(X = x_i, Y = y_j).$$

- **Product rule**:

$$\Pr(X = x_i, Y = y_j) = \Pr(X = x_i | Y = y_j) \cdot \Pr(Y = y_j).$$

$$\Pr(X = x) = \sum_{y \in \mathcal{Y}} \Pr(X = x, Y = y).$$

$$\Pr(X = x, Y = y) = \Pr(X = x | Y = y) \cdot \Pr(Y = y).$$



Bayes' rule

- **Probability mass function (pmf)**:
  $p(x) := \Pr(X = x),\ p(x, y) := \Pr(X = x, Y = y),\ p(x|y) := \Pr(X = x|Y = y).$

- **Bayes' rule/Bayes' Theorem**:

$$p(y|x) = \frac{p(x, y)}{p(x)} = \frac{p(x|y)p(y)}{p(x)} = \frac{p(x|y)p(y)}{\sum_{y' \in \mathcal{Y}} p(x|y')p(y')}.$$

  - 'Invert the casual relationship' between X and Y.
  - Fix x, $p(y|x) \propto p(x|y)p(y)$.
  - posterior $= \dfrac{\text{likelihood} \times \text{prior}}{\text{model evidence}} \propto \text{likelihood} \times \text{prior}.$



Bayes' rule

$$p(y|x) = \frac{p(x|y)p(y)}{\sum_{y' \in \mathcal{Y}} p(x|y')p(y')} \propto p(x|y)p(y) \text{ (when } x \text{ is fixed)}$$

- **Prior:** beliefs or knowledge about y before observing the data x
- **Likelihood:** how likely the observed x is, given a particular value of y
- **Posterior:** updated belief about y after incorporating the prior and the likelihood of the observed data



Continuous random variables

- **Cumulative distribution function (cdf)**: $x \in \mathbb{R} \mapsto F_X(x) = \Pr(X \leq x)$ is differentiable on $\mathbb{R}$.
- **Probability density function (pdf)**:

$$f_X(x) = \frac{\mathrm{d}}{\mathrm{d}x} F_X(x), \quad x \in \mathbb{R}; \quad f_X(x) \geq 0 \quad \forall x \in \mathbb{R} \text{ and } \int_{\mathbb{R}} f_X(x)\mathrm{d}x = 1.$$



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- **Joint probability density function** of continuous r.v.s.:

$$f_{X,Y}(x,y) = \frac{\partial^2}{\partial x \partial y} F_{X,Y}(x,y), (x,y) \in \mathbb{R}^2.$$

- Conditional probability and independence can be defined analogously to the discrete case:

$$\Pr(a < X \leq b,\ c < Y \leq d) = \int_c^d \int_a^b f_{X,Y}(x,y)\mathrm{d}x\mathrm{d}y.$$



Independence

Definition 3.2 (Independence)

Two random variables X and Y are independent if $p_{X,Y}(x,y) = p_X(x)p_Y(y)$ for all $x \in \mathcal{X}$ and $y \in \mathcal{Y}$.

- X and Y are independent if $p_{X|Y}(x|y) = p_X(x)$ for all $x \in \mathcal{X}$ and $y \in \mathcal{Y}$ such that $p_Y(y) > 0$ (what if $p_Y(y) = 0$?).



Expectation and variance

- **Expectation**:
  - Discrete r.v.: $\mathbb{E}X = \sum_{x \in \mathcal{X}} x p_X(x)$.
  - Continuous r.v.: $\mathbb{E}X = \int_{\mathbb{R}} x f_X(x) \mathrm{d}x$.

    If g is a function from the domain of X to $\mathbb{R}$, we can obtain the expectation of $Y = g(X)$ in the same way:

    $$\mathbb{E}Y = \mathbb{E}[g(X)] = \int_{\mathbb{R}} y f_Y(y) \mathrm{d}y = \int_{\mathbb{R}} g(x) f_X(x) \mathrm{d}x.$$

  - Linearity of expectation: if $g(X) = aX + b$ (for constants $a, b \in \mathbb{R}$), then $\mathbb{E}[g(X)] = a\mathbb{E}[X] + b = g(\mathbb{E}[X])$.

- **Variance**:
  - Discrete r.v.: $\mathrm{Var}(X) = \sum_{x \in \mathcal{X}} (x - \mathbb{E}X)^2 p_X(x)$.
  - Continuous r.v.: $\mathrm{Var}(X) = \int_{\mathbb{R}} (x - \mathbb{E}X)^2 f_X(x) \mathrm{d}x$.



Function of random variable

If g is a function from the domain of X to $\mathbb{R}$, we can obtain the expectation of $Y = g(X)$ in the same way.

Proof. I. Discrete case:

$$\mathbb{E}Y = \mathbb{E}[g(X)] = \sum_{y} y p_Y(y) = \sum_{i=1}^{m} g(x) p_X(x).$$



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can be shown as below:

$$\sum_{i=1}^{m} g(x_i) p_X(x_i) = \sum_y \sum_{x:g(x)=y} g(x) \Pr(X = x) = \sum_y \sum_{x:g(x)=y} y \Pr(X = x)$$



Function of random variable

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$$\begin{aligned}\sum_{i=1}^{m} g(x_i) p_X(x_i) &= \sum_y \sum_{x:g(x)=y} g(x) \Pr(X = x) = \sum_y \sum_{x:g(x)=y} y \Pr(X = x) \\ &= \sum_y \sum_{x:g(x)=y} y \Pr(Y = y, X = x) = \sum_y y \sum_{x:g(x)=y} \Pr(Y = y, X = x)\end{aligned}$$



Function of random variable

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can be shown as below:

$$\begin{aligned}
\sum_{i=1}^{m} g(x_i) p_X(x_i) &= \sum_y \sum_{x:g(x)=y} g(x) \Pr(X = x) = \sum_y \sum_{x:g(x)=y} y \Pr(X = x) \\
&= \sum_y \sum_{x:g(x)=y} y \Pr(Y = y, X = x) = \sum_y y \sum_{x:g(x)=y} \Pr(Y = y, X = x) \\
&= \sum_y y \Pr(Y = y) = \sum_y y p_Y(y) = \mathbb{E}Y = \mathbb{E}[g(X)].
\end{aligned}$$



Function of random variable

Proof (continued). II. Continuous case:

$$\mathbb{E}Y = \mathbb{E}[g(X)] = \int_{\mathbb{R}} y f_Y(y) \mathrm{d}y = \int_{\mathbb{R}} g(x) f_X(x) \mathrm{d}x.$$

Refer to Theorem 4.1.1 in the book ' Probability and Statistics'.



Function of random variable

If g is a function from the domain of X to $\mathbb{R}$, we can obtain the expectation of $Y = g(X)$ in the same way.

$$\text{(Discrete case)} \qquad \mathbb{E}Y = \mathbb{E}[g(X)] = \sum_y y p_Y(y) = \sum_{i=1}^{m} g(x) p_X(x),$$

$$\text{(Continuous case)} \qquad \mathbb{E}Y = \mathbb{E}[g(X)] = \int_{\mathbb{R}} y f_Y(y) \mathrm{d}y = \int_{\mathbb{R}} g(x) f_X(x) \mathrm{d}x.$$



Maximum likelihood estimation (MLE)

- In machine learning, we almost always do NOT have access to the underlying distributions that the data are generated from.
- Rather, we have access to sample data and we would like to use it to estimate some parameters.



Maximum likelihood estimation: Gaussian distribution

- If samples in $\mathcal{D} = \{X_1, \ldots, X_m\}$ are independently drawn from the univariate Gaussian distribution

$$f_X\left(x; \mu, \sigma^2\right) = \mathcal{N}\left(x; \mu, \sigma^2\right) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(x-\mu)^2\right) \quad x \in \mathbb{R}, \tag{3.1}$$

  we can use the samples in $\mathcal{D}$ to estimate the parameter vector $\theta = \left(\mu, \sigma^2\right)$.

- Note: notation $f_X\left(x; \mu, \sigma^2\right)$ or $f_X(x; \theta)$ emphasize that the density is parametrized by the parameters $\left(\mu, \sigma^2\right)$ or θ.



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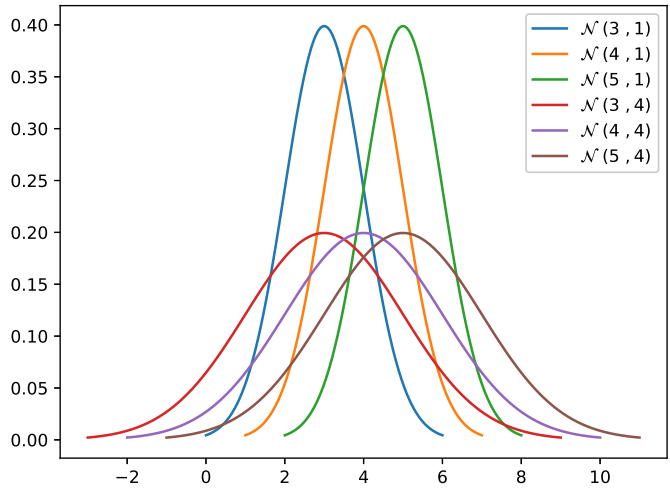
- Note: notation $f_X\left(x; \mu, \sigma^2\right)$ or $f_X(x; \theta)$ emphasize that the density is parametrized by the parameters $\left(\mu, \sigma^2\right)$ or θ.

◎ Maximum likelihood estimates for the mean and variance:

$$\hat{\mu}_{\mathrm{ML}} = \frac{1}{m}\sum_{i=1}^{m} X_i \quad \text{and} \quad \hat{\sigma}^2_{\mathrm{ML}} = \frac{1}{m}\sum_{i=1}^{m}\left(X_i - \hat{\mu}\right)^2. \tag{3.2}$$

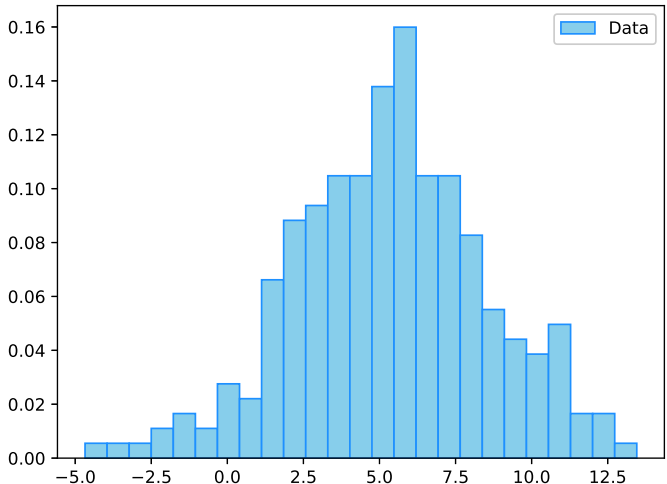


Maximum likelihood estimation: Gaussian distribution



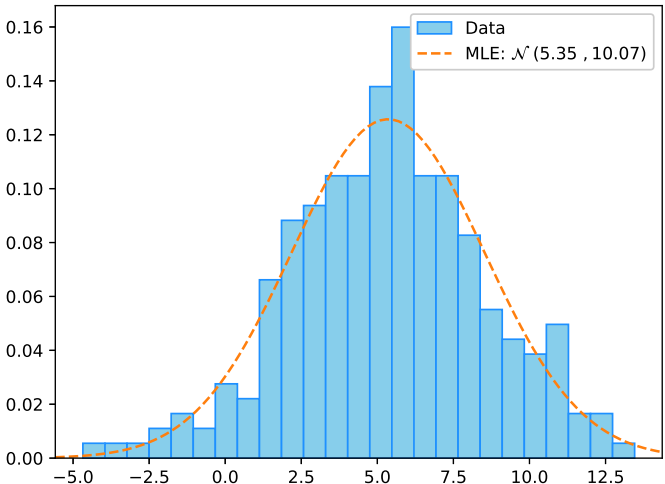


Maximum likelihood estimation: Gaussian distribution



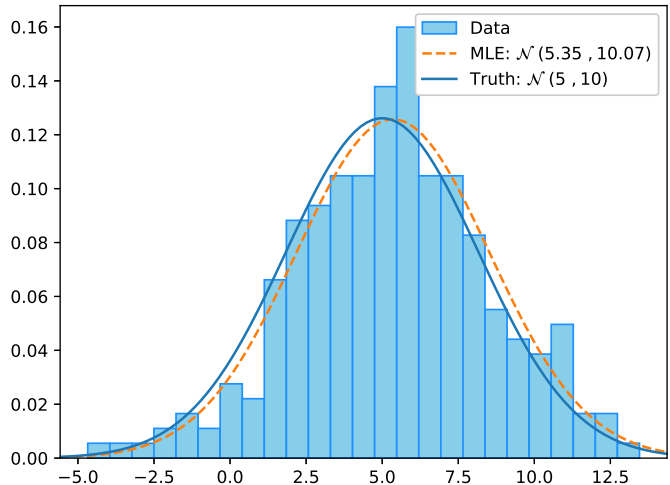


Maximum likelihood estimation: Gaussian distribution





Maximum likelihood estimation: Gaussian distribution





Maximum likelihood estimation: Bernoulli distribution

- Samples in $\mathcal{D} = \{X_1, \ldots, X_m\}$ are generated independently from the Bernoulli (coin toss) distribution

$$p_X(x;\theta) = \begin{cases} 1-\theta & x = 0 \\ \theta & x = 1 \end{cases} \tag{3.3}$$

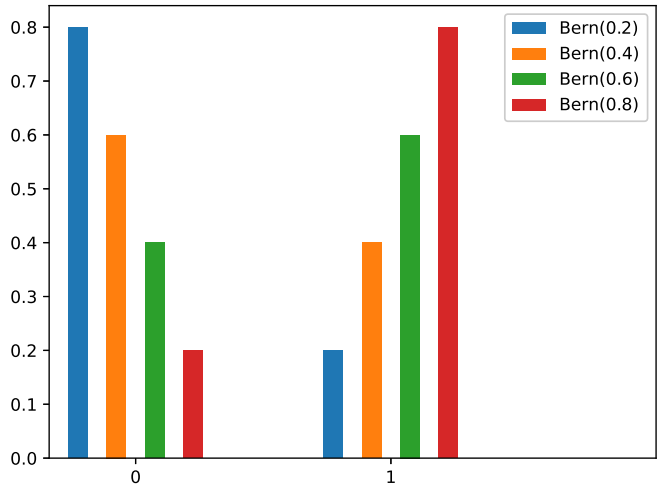
or

$$p_X(x;\theta) = (1-\theta)^{1-x}\theta^x, \quad x \in \{0,1\}. \tag{3.4}$$

- How would we estimate θ from samples?



Maximum likelihood estimation: Bernoulli distribution





Maximum likelihood estimation: Bernoulli distribution

- $M_1 := \sum_{i=1}^{m} X_i$ denote the total number of ones in $\mathcal{D}$. Consider

$$\hat{\theta}_{\mathrm{ML}} = \underset{\theta \in (0,1)}{\arg\max} \prod_{i=1}^{m} p_X(X_i; \theta) \tag{3.5}$$



Maximum likelihood estimation: Bernoulli distribution

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$$= \arg\max_{\theta \in (0,1)} \sum_{i=1}^{m} \log p_X(X_i; \theta) \tag{3.6}$$



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$$= \arg\max_{\theta \in (0,1)} \sum_{i=1}^m \log p_X(X_i; \theta) \tag{3.6}$$

$$= \arg\max_{\theta \in (0,1)} \sum_{i=1}^m [(1 - X_i) \log(1 - \theta) + X_i \log \theta] \tag{3.7}$$

$$= \arg\max_{\theta \in (0,1)} (m - M_1) \log(1 - \theta) + M_1 \log \theta. \tag{3.8}$$



Maximum likelihood estimation: Bernoulli distribution

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- $g(\theta)$ is strictly concave $\Rightarrow$ differentiating and setting to zero yields the (unique) maximum:



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- $g(\theta)$ is strictly concave $\Rightarrow$ differentiating and setting to zero yields the (unique) maximum:

$$g'(\theta) = -\frac{m - M_1}{1-\theta} + \frac{M_1}{\theta}, \tag{3.10}$$

$$\hat{\theta}_{\mathrm{ML}} = \underset{\theta \in (0,1)}{\arg\max} g(\theta) \implies g'(\hat{\theta}_{\mathrm{ML}}) = 0 \implies \hat{\theta}_{\mathrm{ML}} = \frac{M_1}{m}. \tag{3.11}$$



Maximum likelihood estimation: Bernoulli distribution

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$$\hat{\theta}_{\mathrm{ML}} = \underset{\theta \in (0,1)}{\arg\max} g(\theta) \implies g'(\hat{\theta}_{\mathrm{ML}}) = 0 \implies \hat{\theta}_{\mathrm{ML}} = \frac{M_1}{m}. \tag{3.11}$$

- Mean θ is estimated by the empirical mean M_1/m: agrees with common sense!



Maximum likelihood estimation: exponential distribution

- Samples in $\mathcal{D} = \{X_1, \ldots, X_m\}$ are generated independently from the exponential distribution (memory-less)

$$f_X(x;\theta) = \begin{cases} \theta \exp(-\theta x) & x \geq 0 \\ 0 & x < 0 \end{cases}.$$

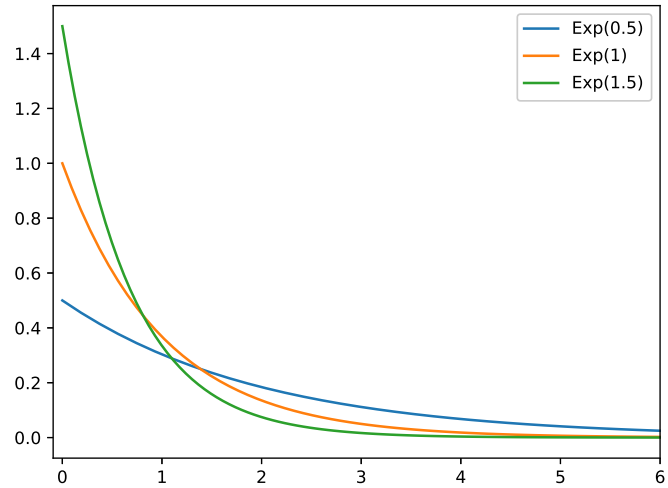
- $\theta > 0$ is the unknown (rate) parameter. In fact, $\mathbb{E}[X] = 1/\theta$.
  - Case: network request arrival time

$$F_X(x;\theta) = \begin{cases} 1 - \exp(-\theta x) & x \geq 0 \\ 0 & x < 0 \end{cases}.$$

- How would we estimate θ from samples?



Maximum likelihood estimation: exponential distribution





Maximum likelihood estimation: exponential distribution

- Consider

$$\hat{\theta}_{\mathrm{ML}} = \arg\max_{\theta>0} \prod_{i=1}^{m} f_X\left(X_i;\theta\right) = \arg\max_{\theta>0} \sum_{i=1}^{m} \log f_X\left(X_i;\theta\right) \tag{3.12}$$



Maximum likelihood estimation: exponential distribution

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$$= \arg\max_{\theta>0} \sum_{i=1}^{m} (\log\theta - \theta X_i) = \arg\max_{\theta>0} m\log\theta - \sum_{i=1}^{m} \theta X_i. \tag{3.13}$$



Maximum likelihood estimation: exponential distribution

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$$\hat{\theta}_{\mathrm{ML}} = \arg\max_{\theta>0} \prod_{i=1}^{m} f_X(X_i;\theta) = \arg\max_{\theta>0} \sum_{i=1}^{m} \log f_X(X_i;\theta) \tag{3.12}$$

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- $g(\theta) = m\log\theta - \sum_{i=1}^{m} \theta X_i$ is strictly concave $\Rightarrow$ differentiating and setting to zero yields the (unique) maximum:



Maximum likelihood estimation: exponential distribution

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- $g(\theta) = m\log\theta - \sum_{i=1}^{m} \theta X_i$ is strictly concave $\Rightarrow$ differentiating and setting to zero yields the (unique) maximum:

$$g'(\theta) = \frac{m}{\hat{\theta}_{\mathrm{ML}}} - \sum_{i=1}^{m} X_i, \tag{3.14}$$

$$g'(\hat{\theta}_{\mathrm{ML}}) = 0 \implies \hat{\theta}_{\mathrm{ML}} = \hat{\theta}_{\mathrm{ML}} = \frac{m}{\sum_{i=1}^{n} X_i} = \left(\frac{1}{m}\sum_{i=1}^{m} X_i\right)^{-1}. \tag{3.15}$$



Maximum likelihood estimation: exponential distribution

- Exponential distribution (memoryless distribution)

$$f_X(x;\theta) = \begin{cases} \theta \exp(-\theta x) & x \geq 0 \\ 0 & x < 0 \end{cases}.$$

- $\mathbb{E}[X] = 1/\theta$, $\hat{\theta}_{\mathrm{ML}} = \left(\frac{1}{m}\sum_{i=1}^{m} X_i\right)^{-1}$.



Maximum likelihood estimation: exponential distribution

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$$f_X(x;\theta) = \begin{cases} \theta \exp(-\theta x) & x \geq 0 \\ 0 & x < 0 \end{cases}.$$

- $\mathbb{E}[X] = 1/\theta$, $\hat{\theta}_{\mathrm{ML}} = \left(\frac{1}{m}\sum_{i=1}^{m} X_i\right)^{-1}$.

◎ $\frac{1}{m}\sum_{i=1}^{m} X_i$ is the empirical mean:
it seems plausible that $m/\left(\sum_{i=1}^{m} X_i\right)$ is a "good" estimate of θ.

- Why would it not be "good"?



Maximum likelihood estimation

$$\hat{\theta}_{\mathrm{ML}} = \underset{\theta \in (0,1)}{\arg\max}\ p_{X_1, x_2, \ldots, x_m}\left(X_1, X_2, \ldots, X_m; \theta\right)$$



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- Maximum: seeks the set of parameters θ that maximizes this likelihood function, i.e., makes the observed dataset as "likely" as possible under the model



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- Likelihood: probability of observing the dataset $\mathcal{D}$ given a set of parameters θ
- Maximum: seeks the set of parameters θ that maximizes this likelihood function, i.e., makes the observed dataset as "likely" as possible under the model
- Concept of MLE: linear regression, logistic regression, ...



Maximum likelihood estimation (MLE)

- **Consistency.** as the sample size increases to infinity, the estimator will converge to the true parameter value: $\hat{\theta}_{\mathrm{ML}} \xrightarrow{p} \theta$ as $n \to \infty$.
  - We say $X_n \xrightarrow{p} X$ as $n \to \infty$ if

$$\lim_{n \to \infty} \Pr(|X_n - X| > \varepsilon) = 0 \ \forall \varepsilon > 0.$$



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- **Unbiasedness.** MLE is not necessarily unbiased, but in certain cases, it can be unbiased.
  - We say an estimator of a given parameter is unbiased if its expected value is equal to the true value of the parameter.



Maximum likelihood estimation (MLE)

- **Efficiency.** MLE is asymptotically efficient in large samples, i.e., it achieves the lowest possible variance among all unbiased estimators.
  - Cramér-Rao Lower Bound (CRLB): theoretical lower bound for the variance of any unbiased estimator.



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- **Asymptotic Normality.** MLE is asymptotically normal: $\sqrt{n}(\hat{\theta}_{\mathrm{ML}} - \theta) \xrightarrow{d} \mathcal{N}(0, I(\theta)^{-1})$.
  - $I(\theta)$: Fisher information of θ (larger information $\Longrightarrow$ smaller variance).
  - We say $X_n \xrightarrow{d} X$ as $n \to \infty$ if

$$\lim_{n\to\infty} F_{X_n}(x) = F_X(x) \ \forall x \text{ at which } F_X(x) \text{ is continuous.}$$



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- **Model Sensitivity.** MLE is sensitive to model assumptions, and incorrect assumptions can lead to biased or inconsistent estimates.



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Outline

1. Logistics
2. Introduction to Machine Learning
   - Definition and taxonomy
3. Mathematical tools
   - Set and function
   - Probability and estimation
   - Systems of linear equations



Systems of linear equations

Definition 3.3 (Rules)

A vector space over the reals consists of a set $\mathcal{V}$, a vector sum operation $+ : \mathcal{V} \times \mathcal{V} \to \mathcal{V}$ and a scalar multiplication operation $\cdot : \mathbb{R} \times \mathcal{V} \to \mathcal{V}$ satisfying the following properties.

- Commutativity: $\boldsymbol{x} + \boldsymbol{y} = \boldsymbol{y} + \boldsymbol{x}$ for all $\boldsymbol{x}, \boldsymbol{y} \in \mathcal{V}$;
- Associativity: $(\boldsymbol{x} + \boldsymbol{y}) + \boldsymbol{z} = \boldsymbol{x} + (\boldsymbol{y} + \boldsymbol{z})$ for all $\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathcal{V}$;
- Identity element of addition: $\boldsymbol{x} + \boldsymbol{0} = \boldsymbol{x}$ for all $\boldsymbol{x} \in \mathcal{V}$;
- Inverse element of addition: For every $\boldsymbol{x} \in \mathcal{V}$, there exists an element $-x \in \mathcal{V}$ such that $\boldsymbol{x} + (-\boldsymbol{x}) = \boldsymbol{0}$;
- Associativity of scalar multiplication: For all $a, b \in \mathbb{R}$ and $x \in \mathcal{V}$, $a(b\boldsymbol{x}) = (ab)\boldsymbol{x}$;
- Identity element of scalar multiplication: $1\boldsymbol{x} = \boldsymbol{x}$ for $\boldsymbol{x} \in \mathcal{V}$;
- Distributivity of scalar multiplication with respect to vector addition: $a(\boldsymbol{x} + \boldsymbol{y}) = a\boldsymbol{x} + a\boldsymbol{y}$ for all $a \in \mathbb{R}$ and $\boldsymbol{x}, \boldsymbol{y} \in \mathcal{V}$;
- Distributivity of scalar multiplication with respect to addition in $\mathbb{R}$: $(a + b)\boldsymbol{x} = a\boldsymbol{x} + b\boldsymbol{x}$ for all $a, b \in \mathbb{R}$ and $\boldsymbol{x} \in \mathcal{V}$.



Vector

Definition 3.4 (Linear independence)

A set of vectors $\{\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_k\}$ from a vector space $\mathcal{V}$ is linearly independent if

$$\beta_1 \boldsymbol{x}_1 + \beta_2 \boldsymbol{x}_2 + \ldots + \beta_k \boldsymbol{x}_k = \mathbf{0} \tag{3.16}$$

implies that $\beta_1 = \beta_2 = \ldots = \beta_k = 0$.

- $\{\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_k\}$ are linearly independent $\Longleftrightarrow$ no vector $\boldsymbol{x}_i$ can be expressed as a linear combination of the other vectors $\{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_{i-1}, \boldsymbol{x}_{i+1}, \ldots, \boldsymbol{x}_k\}$.



Matrix

Definition 3.5 (Basis of vector space)

A set of vectors $\{\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_k\}$ is a basis for a vector space $\mathcal{V}$ if

- $\mathcal{V} = \text{span}\{\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_k\}$;
- $\{\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_k\}$ is a **linearly independent set** of vectors.

Equivalently, every $\boldsymbol{x} \in \mathcal{V}$ can be uniquely written as $\sum_{i=1}^{k} \beta_i \boldsymbol{x}_i$ for some $\{\beta_i\}_{i=1}^{k} \subset \mathbb{R}$. The number of vectors in any basis of V is called the **dimension** of $\mathcal{V}$, written as $\dim(\mathcal{V})$.

Definition 3.6 (nullspace and range of matrix)

The **nullspace (also called kernel)** of a matrix $\boldsymbol{A} \in \mathbb{R}^{m \times d}$ is defined as

$$\mathcal{N}(\boldsymbol{A}) = \{\boldsymbol{x} \in \mathbb{R}^d : \boldsymbol{A}\boldsymbol{x} = 0\}. \tag{3.17}$$

The range or column space of $\boldsymbol{A}$ is defined as

$$\mathcal{R}(\boldsymbol{A}) = \{\boldsymbol{A}\boldsymbol{x} : \boldsymbol{x} \in \mathbb{R}^f\} \subset \mathbb{R}^m. \tag{3.18}$$



Matrix

Definition 3.7 (Rank of matrix)

The **column rank or rank** of a matrix $\boldsymbol{A} \in \boldsymbol{R}^{m \times d}$ is defined as

$$\operatorname{rank}(\boldsymbol{A}) = \dim(\mathcal{R}(\boldsymbol{A})). \tag{3.19}$$

In other words, the column rank of $\boldsymbol{A}$ is the dimension of the column space of $\boldsymbol{A}$.



Matrix

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In other words, the column rank of $\boldsymbol{A}$ is the dimension of the column space of $\boldsymbol{A}$.

- **Rank-nullity**: $\operatorname{rank}(\boldsymbol{A}) + \dim(\mathcal{N}(\boldsymbol{A})) = d$.
- It is always true that $\operatorname{rank}(\boldsymbol{A}) \leq \min\{m, d\}$.
- A matrix is full rank if $\operatorname{rank}(\boldsymbol{A}) = \min\{m, d\}$.
- A matrix is full column rank (resp. full row rank) if the set of columns (resp. rows) of the matrix is linearly independent.
- If the matrix A is square (i.e., $m = d$) and it is full rank, then the inverse $\boldsymbol{A}^{-1}$ exists.



Solutions to linear systems

Solve systems of equations of the form ($\underline{x}_i,\ 1 \le i \le d$: columns of $\boldsymbol{X}$)

$$\boldsymbol{X}\boldsymbol{w} = \boldsymbol{y} \quad \text{or} \quad [\underline{x}_1\ \underline{x}_2\ \dots\ \underline{x}_d] \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_d \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_d \end{bmatrix}, \text{ where } \underline{x}_i = \begin{bmatrix} x_{1,i} \\ x_{2,i} \\ \vdots \\ x_{m,i} \end{bmatrix}. \tag{3.20}$$

Matrix $\boldsymbol{X} \in \mathbb{R}^{m \times d}$ and vector $\boldsymbol{y} \in \mathbb{R}^m$ are given and $\boldsymbol{w} \in \mathbb{R}^d$ is to be found.



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- $m = d$: if matrix $\boldsymbol{X}$ is square and full rank, $\boldsymbol{X}^{-1}$ exists $\Longrightarrow$ solve for $\boldsymbol{w}$ by simple matrix inversion and multiplication $\boldsymbol{w} = \boldsymbol{X}^{-1}\boldsymbol{y}$.
- $m \neq d$: existence and uniqueness of solutions to the linear system?

Augmented matrix $\tilde{\boldsymbol{X}} = [\boldsymbol{X}\boldsymbol{y}] \in \mathbb{R}^{m \times (d+1)}$: $\operatorname{rank}(\boldsymbol{X}) \le \operatorname{rank}(\tilde{\boldsymbol{X}})$.



Solutions to linear systems

◎ **Method**: apply Guassian-elimination method to the augmented matrix $\tilde{\boldsymbol{X}}$.

Theorem 3.8 (Rouché-Capelli Theorem)

- *The system in* (3.20) *admits a unique solution if and only if* $\operatorname{rank}(\boldsymbol{X}) = \operatorname{rank}(\tilde{\boldsymbol{X}}) = d$;
- *The system in* (3.20) *has no solution if and only if* $\operatorname{rank}(\boldsymbol{X}) < \operatorname{rank}(\tilde{\boldsymbol{X}})$;
- *The system in* (3.20) *has infinitely many solutions if and only if* $\operatorname{rank}(\boldsymbol{X}) = \operatorname{rank}(\tilde{\boldsymbol{X}}) < d$.



Take-away



Take-away

1. Logistics
2. Introduction to Machine Learning
   - Definition and taxonomy
3. Mathematical tools
   - Set and function
   - Probability and estimation
   - Systems of linear equations



Syllabus

Lec #	Topic	Lecturer
1	**Introduction + course Info**	Zixin
2-4	Supervised learning: regression and classification	Zixin
5	SVM (supervised learning) + Feature engineering	Zixin
6	Model evaluation and choice	Zixin
7	Ensemble learning (supervised learning) + Feature selection	Weikai
	Midterm (28 March): mark it on calendar!	
8-9	Unsupervised learning: clustering	Weikai
10	Dimension reduction	Weikai
11	Markov and graphical models	Weikai
12	Active learning + Reinforcement learning	Weikai
13	Project presentation	Both
	Project report submission (TBD)	
	Final exam (TBD)	



Academic calendar

Week	S	M	T	W	T	F	S
	January, 2026						
					1	2	3
	4	5	6	7	8	**9**	10
	11	12	13	14	**15**	16	17
1	18	19	20	21	**22**	**23**	**24**
1	25	**26**	**27**	**28**	**29**	**30**	**31**
	February						
2	1	**2**	**3**	**4**	**5**	**6**	**7**
3	8	**9**	**10**	**11**	**12**	**13**	**14**
	15	16	17	18	19	20	21
3	22	23	**24**	**25**	**26**	**27**	**28**
	March						
4	1	**2**	**3**	**4**	5	6	7
5	8	9	10	11	12	13	14
6	15	16	17	18	19	20	21
7	22	23	24	25	26	27	28
8	29	30	31				
	April						
8				1	2	3	4
9	5	6	7	8	9	10	11
10	12	13	14	15	16	17	18
11	19	20	21	22	23	24	25
12	26	27	28	29	30		
	May						
						1	2
13	3	4	5	6	7	8	9
13	10	**11**	**12**	**13**	**14**	**15**	**16**
	17	**18**	**19**	**20**	**21**	**22**	**23**
	24	**25**	**26**	**27**	**28**	**29**	30
	31						

Events		General Holidays	
		1-3	*New Year's Day*
9	Validation Period Starts - All UG/PG Students		
15	Class Enrollment Starts - All UG/PG Students		
22	Last day of Winter Term		
23	Spring Term commences		
23/1-5/2	Add/Drop Period		
11-23	Winter Break for UG students (no centrally scheduled PG classes)	15-23	*Spring Festival*
28	Make-up day for Monday classes		
6/2-4/3	Extended Drop Period - For PG Courses Only		
		4-6	*Qingming Festival*
		1-5	*Labor Day*
11	Last day of Spring Term classes		
12-16	Study Break		
18-29	Spring Term Examinations		
29	Last day of Spring Term		



Try Canvas: https://hkust-gz.instructure.com/courses/

- Quizzes
- Syllabus
- Announcements, Modules
- Discussions, Chat
- Grades





Taxonomy of Machine Learning (A Simplistic View)

- What type of data?
  - **Supervised learning** - labeled data: e.g., prediction
  - Semi-supervised learning
  - **Unsupervised learning** - unlabeled data: e.g., clustering
  - **Reinforcement learning** - environment feedback: e.g., multi-armed bandit

- When do we collect data?
  - **Offline learning**
  - **Online learning**



Mathematical tools

- Set and function
  - Set
  - Function: differentiable, (strictly) convex/concave, linear and affine
  - Local and global extrema, partial derivative and gradient vector



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- Probability and estimation
  - Discrete and continuous random variables, event
  - Sum rule and product rule
  - Bayes' rule
  - Independence, expectation and variance



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    Find the parameter set that maximizes the probability the given dataset is collected
  - Exponential distribution: memory-less



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- Systems of linear equations
  - Vector: linear independence
  - Matrix: rank, kernel, range, dimension, invertible
  - Solution to linear system: existence and uniqueness, Gaussian-elimination method



Thanks for listening

1. Tell us your question/feedback via the QR code.
2. Lab reminder: Th 7-8PM[LA01] / 6-7PM[LA02]



⊙ Slides credit: some slides are adapted from (alphabetical order)
Haiyun He (IoT), Tengyu Ma and Chris Re (Stanford) and Vincent Y. F. Tan (NUS).



References

A. L. Samuel. Some studies in machine learning using the game of checkers. IBM Journal of Research and Development, 3(3):210–229, 1959. doi: 10.1147/rd.33.0210.

